

Grand Challenge of Safety Critical Human Motion Generation

Bonus Questions, 2025

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Note: This is a bonus project that will contribute to the overall course points. For instance, if your accumulated course score is 81 and you have 9.5 points from this challenge, your overall course score will be 90.5, and you will be directly upgraded to an A grade.

It's encouraged to collaborate with your original team members on the project. However, you can also work independently, with a subset of your team, or even with individuals from other teams. **You must clearly document and acknowledge each person's contributions to the project in accordance with the honor code at the beginning of your report.** In a team setting, each individual's contribution will be equally valued and contribute to the overall team score.

Simulating human motion in a simulator is essential for evaluating and enhancing autonomous systems, ensuring they can interact safely with humans. By replicating realistic human dynamics, simulations enable testing and optimization without exposing people to danger. Ultimately, these simulations serve as a foundation for developing safer, more intuitive, and adaptive autonomy systems. Realistic human motion requires two key components: accurately replicating human movement and integrating it into a simulation environment. This bonus assignment is divided into two parts, each focused on practicing one of these components. The first part involves estimating motion from video, while the second part entails rendering the estimated motion within the CARLA environment

1 From Video to SMPLH (10 points)

In this part, you will use an open-source model to extract human motion from a video and save the SMPLH data, which is then rendered into a video showing the human mesh.

1.1 Run the SMPLH data rendering (2 points)

The code, smplh model checkpoint, and open-source data can be found [here](#). When running the script `render_smplh.py`, you will see a video that shows the mesh human moving, as shown in Figure 1. Make sure you put the script, SMPLH model folder, and the data folder under the same path.

For this step, submit a link to the saved video.



Figure 1: SMPL data visualized in mesh.

1.2 Run the model to collect SMPLH data (4 points)

WHAM: Reconstructing World-grounded Humans with Accurate 3D Motion is a human motion reconstruction method with SOTA performance. From the GitHub page, you can find colab link to run their demo. Run the demo with an extra `--save-pkl` and after running the demo, you can find a `wham_output.pkl` file saved along with the video. This file contains the smplh data, including pose and betas. The video shows the original frames with a mesh mask, as shown in Figure 2.

For this step, submit a link to the saved pkl data file and the saved video with mesh.



Figure 2: Caption

1.3 Render the saved SMPLH data (4 points)

Modify the `render_smplh.py` to load the `wham_output.pkl` file and render the video.

For this step, submit a link to the saved video.

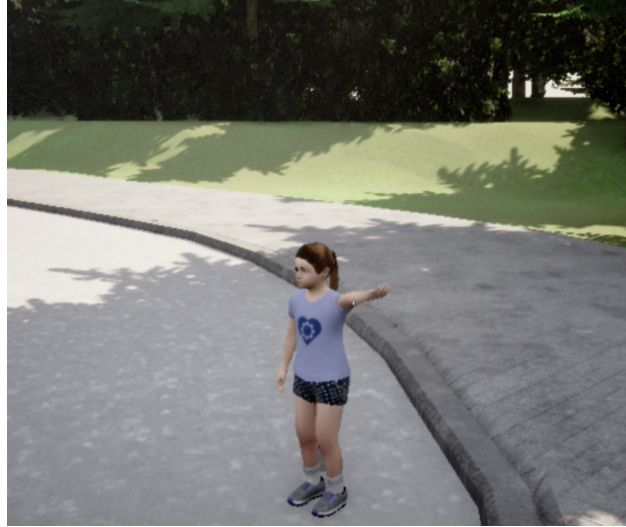


Figure 3: Caption

2 From SMPLH to Carla (5 points)

In this part, you will use render the SMPLH data in the Carla environment. The starter code can be found [here](#). The starter code may be updated later. If any updates are made, there will be an announcement on CampusWire. Hints and more details will also be posted on CampusWire.

2.1 Run the skeleton controller (2 points)

Start Carla as *challenge-2*. Run the `test_controller.py`, and save the video for one mapping option. The rendering result are like Figure 3.

For this step, submit a link to the saved videos.

2.2 Modify the skeleton controller to render the motion correctly (3 points)

Create a mapping from SMPL to CARLA to have smooth and correct human motions.

For this step, submit code of your mapping function and a video of rendering human motion.

3 Reference

- [1] Loper, M., Mahmood, N., Romero, J., Pons-Moll, G., & Black, M. J. (2023). SMPL: A skinned multi-person linear model. In *Seminal Graphics Papers: Pushing the Boundaries, Volume 2* (pp. 851-866).
- [2] Pavlakos, G., Choutas, V., Ghorbani, N., Bolkart, T., Osman, A. A., Tzionas, D., & Black, M. J. (2019). Expressive body capture: 3d hands, face, and body from a single

- image. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 10975-10985).
- [3] Shin, S., Kim, J., Halilaj, E., & Black, M. J. (2024). Wham: Reconstructing world-grounded humans with accurate 3d motion. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 2070-2080).
- [4] WHAM repo: <https://github.com/yohanshin/WHAM?tab=readme-ov-file>